

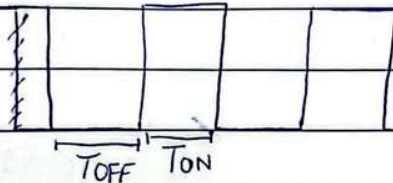
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CH # 7 SWITCHING POWER SUPPLIES (DC to DC)

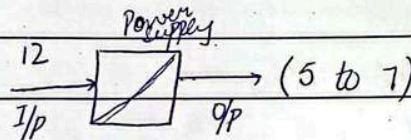
CASE-I:

- $V_{in} \rightarrow \text{Constant}$
 $V_{out} \rightarrow \text{Constant}$ } fixed, & also D-fixed



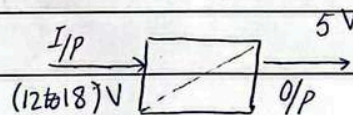
CASE-II:

- $V_{in} \rightarrow \text{Constant/fixed}$
 $V_{out} \rightarrow \text{Variable}$
 $\Rightarrow D \rightarrow \text{Variable}$



CASE-III:

- $V_{in} \rightarrow \text{variable}$
 $V_{out} \rightarrow \text{fixed}$
 $D \rightarrow \text{variable}$



CASE-IV

- $V_{in} \rightarrow \text{variable}$
 $V_{out} \rightarrow \text{variable}$

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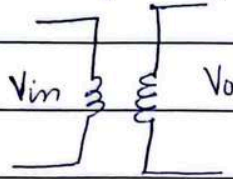
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BUCK REGULATORS / CONVERTERS

($V_{in} > V_o$)

- Input is high, o/p. is low

eg#

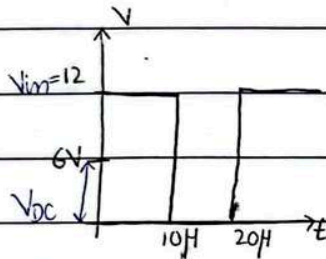


Step down transformer

* $V_{out} = D \times V_{in}$

- ideally $D = 1$

1)



$$V_{avg} = 12 \times \frac{10}{20} = 6V$$

$\nearrow T_{ON}$
 $\searrow T_{ON+OFF}$

2) $V_{avg} = 12 \times (5/20) = 3V$

3) $V_{in} = 7V$

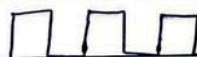
$V_{out} = 6V$

$$D = \frac{V_{out}}{V_{in}} = \frac{6}{7} = 0.85$$

- Switch driver is used to control switch.
- Schottky diode in parallel with switch - used to prevent switch.
(Zener)
- O/p of switch is rectangle, filter required.

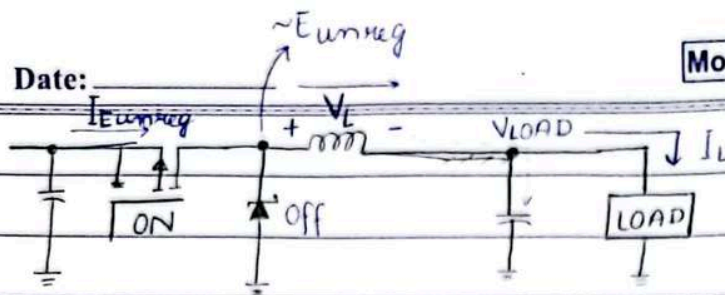
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PWM



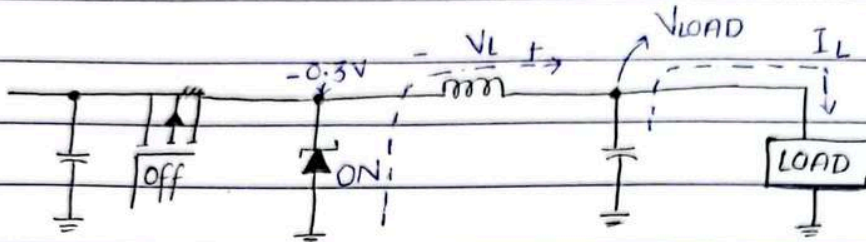
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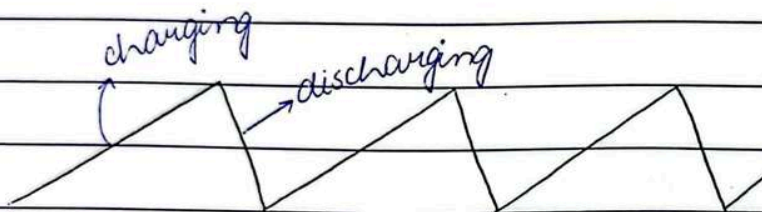
charge part of the cycle

Buck
regulator
operation



Discharge part of the cycle

* V_L & V_{LOAD}
are different



o/p graph

*
current rating $> 1.3 I_{DC, LOAD}$
PIV rating $> 1.3 E_{unreg, max}$

$$V_L = L \frac{\Delta i}{\Delta t}$$

* inductor - opposes
change in current

$$\begin{aligned} V_L &= E_{unreg} - V_{LOAD} \\ &= V_i - V_o \\ &= V_i - DV_i \\ &= (1 - D)V_i \end{aligned}$$

Analysis $\rightarrow$ calculate Δi
Design $\rightarrow$ calculate L

Δi - AC current capacitor
will entertain.

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* ripple current - depends on inductor

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$$\Delta t = t_{ON} = DT$$

(duty cycle is set by the relationship of i/p to o/p)
 $\Rightarrow D = V_{LOAD} / E_{unreg}$

* Select $\Delta i = 0.25 I_{DC}$ to $0.7 I_{DC}$ (Ripple current)

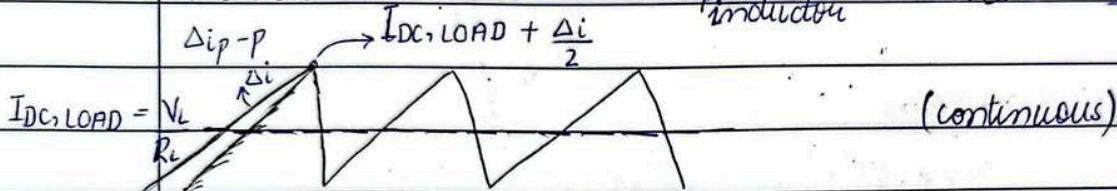
→ * Load current will be smooth current.

* [→ At or below 200mA DC, allow longer ripple current.

→ Above 1A DC, restrict ripple current to 25%.

→ In analysis, ripple current is set by inductor.

* Inductor max current = $i_{p_inductor} = I_{DC, LOAD} + \frac{\Delta i}{2}$



EXAMPLE # 7.2

$$V_{in} = E_{unreg} = 12V, \quad f = 100kHz \Rightarrow T = 1/100k = 0.01ms$$

$$V_{LOAD} = V_o = 5V, \quad L = 220 \mu H$$

$$R_L = 10 \Omega$$

i) Duty cycle = $D = ?$, ii) on time = $\Delta t = ?$

iii) ripple current = $\Delta i = ?$, iv) avg. Inductor Current = $I_{DC, LOAD} = ?$

v) Peak inductor current = $I_{L, Peak} = ?$

↑ voltage across inductor

$$V_L = E_{unreg} - V_{LOAD} \\ = 12 - 5 = 7V$$

$$i) \quad D = \frac{V_{LOAD}}{E_{unreg}} = \frac{V_o}{V_{in}} = \frac{5}{12} = 42\%$$

$$ii) \quad \Delta t = D \times T = 0.42 (0.01ms) \\ = 4.2 \mu s$$

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POWER ELECTRONICS

CHAPTER # 7

EXAMPLE # 7.2 (Remaining)

iii)

$\Delta i = ?$

$V_L = 12 - 5 \quad \text{OR} \quad V_L = (1-D)V_{in}$

$$V_L = L \frac{\Delta i}{\Delta t}$$

$$\therefore V_L = 7 \text{ V} \Rightarrow (1-D)V_{in} = L \frac{\Delta i}{\Delta t}$$

$$\Delta i = \frac{(1-0.42)(12)(4.2 \mu)}{220 \mu}$$

$$\Delta i = \frac{V_L \Delta t}{L} = \frac{7(4.2 \mu)}{220 \mu}$$

$$\Delta i = 133 \text{ mA}$$

$$\Delta i = 134 \text{ mA}$$

* To verify:

iv)

$$I_{DC,LOAD} = \frac{V_{LOAD}}{R_L} = \frac{5 \text{ V}_{DC}}{10} = 500 \text{ mA}_{DC}$$

$$\frac{\Delta i}{I_{DC,LOAD}} = \frac{134 \text{ mA}_{PP}}{500 \text{ mA}_{DC}}$$

$$= 0.27 \Rightarrow 27\%$$

 $\Rightarrow$ close to 25% $\rightarrow$ (reasonable)

v)

$$I_{L,peak} = \frac{I_{DC,LOAD} + \Delta i}{2} = \frac{500 \text{ mA}_{DC} + 134 \text{ mA}_{PP}}{2} = 567 \text{ mA}$$

OR using $\Delta i = 133 \text{ mA}$

$$\Rightarrow I_{L,peak} = \frac{500 \text{ mA} + 133 \text{ mA}}{2} = 565 \text{ mA}$$

 $\nearrow$ (o/p smaller than I_{IP})

7.1 *

BUCK REGULATORS: (Remaining)

 $\rightarrow$ A key parameter of for C_m is its equivalent series resistance. $\rightarrow$ Large capacitors, C_m & C_{out} may have hundred milliohms of internal resistance. $\rightarrow$ Current flowing off and onto these capacitors produces a voltage drop & forces capacitor to dissipate power (producing heat).

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$$I_{RMS, Cin} > 0.5 I_{DC, LOAD} \quad (I_p \text{ capacitor ratings})$$

$$V_{Cin} > 1.5 E_{unreg, max}$$

→ * Any capacitor & above 100 μF should work.

$$\Delta V_{PP, LOAD} = \Delta i_{pp, inductor} \times ESR_{cout} \quad (o/p \text{ capacitor's specifications})$$

$$I_{RMS, cout} = \frac{\Delta i_{pp, inductor}}{\sqrt{3}}$$

$$C \approx \frac{500 \mu F}{A} \times I_{DC, LOAD}$$

EXAMPLE # 7.3

$$I_{DC, LOAD} = 500 \text{ mA DC}$$

$$\Delta i_{pp} = 134 \text{ mA PP}$$

$$D = 0.42$$

i) $I_{RMS} = ?$ (using trapezoid equation)

ii) $I_{RMS} = ?$ (using simple rectangle with $I_p = I_{DC, LOAD}$)

i) For trapezoid:

$$I_{RMS} = \sqrt{\frac{D}{3} (I_p^2 + I_p I_{min} + I_{min}^2)} \quad (i)$$

where,

$$I_p = I_{DC} + \frac{\Delta i_{pp}}{2}$$

$$\& \quad I_{min} = I_{DC} - \frac{\Delta i_{pp}}{2}$$

$$= \frac{500 \text{ mA} + 134 \text{ mA}}{2}$$

&

$$= \frac{500 \text{ mA} - 134 \text{ mA}}{2}$$

$$I_p = 567 \text{ mA PP}$$

&

$$I_{min} = 433 \text{ mA PP}$$

$$\Rightarrow \text{ii) } I_{RMS} = \sqrt{\frac{0.42}{3} [(567 \text{ mA})^2 + (567 \text{ mA})(433 \text{ mA}) + (433 \text{ mA})^2]}$$

$$= 325 \text{ mA RMS}$$

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ii) For rectangle (with flat top of 500mA)

$$I_{RMS} = \sqrt{D \times I_p} = \sqrt{0.42 \times 500 \text{mA}}$$

$$= 324 \text{mA}_{RMS}$$

$$\Delta i = \frac{\Delta i_{pp}}{I_{DC, LOAD}} \times 100 = \frac{134}{500} \times 100$$

$$\Delta i = 26.8\% = 27\%$$

7/4

EXAMPLE #

$$I_{DC, LOAD} = 500 \text{mA}, \Delta i_{pp} = 134 \text{mA},$$

$$D = 0.42 (R_{ON} = 0.6 \Omega \text{ at } \approx 150^\circ\text{C})$$

i) Power dissipation of IRF9530 = ?

$$P_Q = D \times I_{DC, LOAD}^2 \times R_{ON}$$

$$= 0.42 \times (500 \text{mA})^2 \times 0.6$$

$$= 63 \text{mW}$$

ii) Power dissipation by Schottky diode = ?

$$P_{DIODE} = D \times V_{ON} \times I_{DC, LOAD}$$

$$= 0.42 \times 0.3 \times 500 \text{mA}$$

$$= 63 \text{mW}$$

iii) $P_L = ?$, $R_L = 10 \Omega$

$$P_L = I_{DC, LOAD}^2 \times R_L$$

$$= (500 \text{mA})^2 (10)$$

$$= 2.5 \text{W}$$

(No effect of load)
[Π]^x (continuous)

iv) efficiency = $\eta = ?$

$$\eta = \frac{P_L}{P_T} \times 100 = \frac{2.5}{63 \text{m} \times 63 \text{m} \times 2.5} \times 100$$

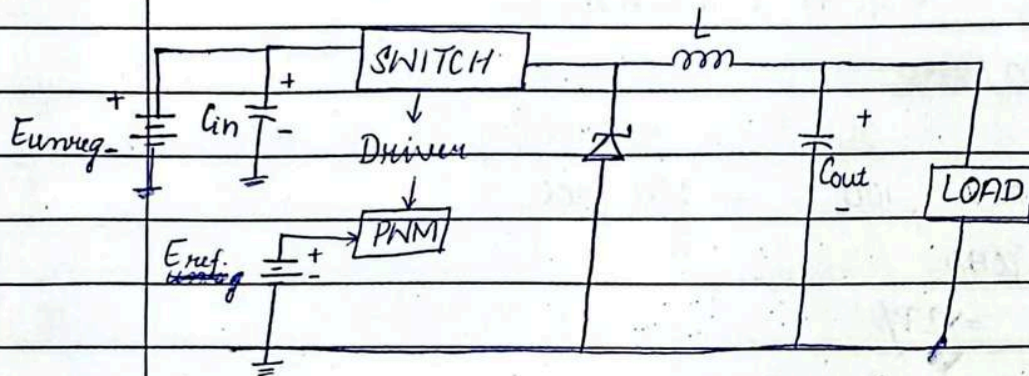
when I_p is known
 $\Rightarrow \eta = \frac{\text{Input} - \text{losses}}{\text{input}}$

$$\eta = 95.2\%$$

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BUCK REGULATOR (LM 2959)



- $P_Q = D \times I_{DC, LOAD} \times V_{sat, max}$
- $P_{IC} = I_{quise} \times E_{unreg}$
- Total Power dissipation = $P_Q + P_{IC}$

EXAMPLE

convert unregulated input voltage of 12V to 5V_{DC}.

$R = 10\Omega$, $T_A = 80^\circ C$, $f = 150kHz$, $V_{ref} = 1.23V$

Sol.

Designing Buck regulator.

$L = ?$

$$V_L = L \frac{\Delta i}{\Delta t}$$

$$V_L = 12 - 5 = 7V$$

$$I_{DC, LOAD} = \frac{5}{10} = 0.5A = 500mA$$

$$\Delta i_{pp} = 0.25 \times 500mA = 125mA$$

$$T = \frac{1}{f} = \frac{1}{150k} = 6.7\mu s$$

$$\begin{aligned} \Delta t &= D \times T \\ &= 0.42 \times 6.7\mu s \\ &= 2.8\mu s \end{aligned}$$

LM2595 specifications

$$V_{sat(max)} = 1V_p$$

$$\theta_{JC} = 2^\circ C/W$$

$$\theta_{JA(bare)} = 50^\circ C/W$$

Select

$$\Delta i = 0.25 \text{ to } 0.7 I_{DC}$$

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$$V_o = DV_{in} \Rightarrow \frac{5}{12} = D$$

$$\Rightarrow L = 157 \mu H \approx 220 \mu H$$

$$(V_L = L \Delta i / \Delta t) \quad \text{125m}$$

$$I_{DC, LOAD} = 500 \text{ mA}$$

$$I_p = \frac{500 + 125}{2} = 563 \text{ mA}$$

* $C_{out} = 250 \mu F$

$$C_{in} > 100 \mu F$$

$$I_{Cin} > 0.5 \times I_{DC, LOAD} = 250 \text{ mA}$$

$$V_{Cin} \geq 1.5 E_{unreg} = 1.5 \times 12 = 18 \text{ V}$$

$$C_{out} \approx 500 \mu F/A \times I_{DC, LOAD}$$

$$P_T = P_Q + P_{IC}$$

$$= (10 \text{ m} \times 12) + (0.42 \times 500 \text{ m} \times 1)$$

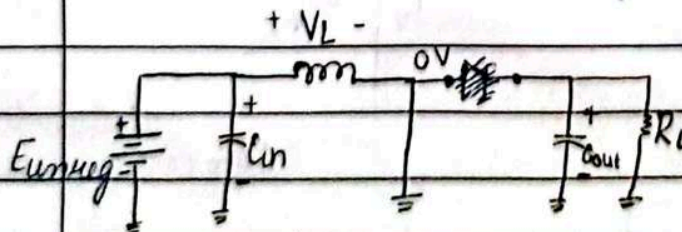
$$= 0.33 \text{ W}$$

$$\theta_{SA} = \frac{150 - 80}{0.33} - 2 - 0.2$$

$$= 210^\circ \text{C/W}$$

* 14/4 BOOST REGULATORS / CONVERTERS

- Low side instead of high side.
- Transistor OFF, diode ON and vice versa.
- Transistor ON: $V_L = E_{unreg} = L \frac{\Delta i}{\Delta t}$ (when charging)



when transistor is ON/
diode is OFF

$$V_L = E_{unreg} - 0$$

$$V_L = E_{unreg} = L \frac{\Delta i}{\Delta t}$$

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current - inductor - ramp
capacitor - constant.

$$\star \Delta i_{pp} - \begin{cases} \text{for buck} = 0.25 \times I_{DC, \text{LOAD}} \\ \text{for boost} = 0.25 \times I_{DC, I/P} \end{cases}$$

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- Boost regulators are 90% efficient.
- * $P_{LOAD} = P_{in} \times 0.9$ OR $V_{DC, \text{LOAD}} \times I_{DC, \text{LOAD}} = 0.9 E_{unreg} \times I_{DC, I/P}$ (i)

→ When transistor is ON:

ANALYSIS:

$$\Delta i = \frac{E_{unreg} \Delta t}{L}$$

DESIGN:

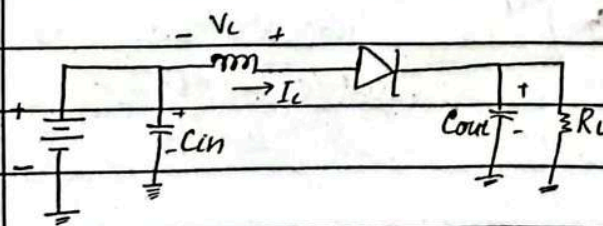
- $\Delta i = 25\%$ of $I_{DC, I/P}$
- $L = \frac{E_{unreg} \Delta t}{\Delta i}$
- $\Delta t = D \times T$
- $I_{L, \text{PEAK}} = I_L + \frac{\Delta i}{2}$ L - inductor
- $V_{LOAD} = E_{unreg} + V_L$
- ⇒ (i) $I_{DC, I/P} = \frac{V_{DC, \text{LOAD}} \times I_{DC, \text{LOAD}}}{0.9 E_{unreg}}$

$$I_L = \frac{V_{DC, \text{LOAD}} \times I_{DC, \text{LOAD}}}{0.9 E_{unreg}}$$

$$\Rightarrow \star I_{DC, I/P} = I_L$$

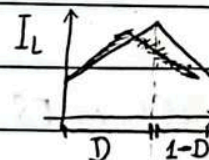
$$I_{avg, L} = \frac{V_{DC, \text{LOAD}} \times I_{DC, \text{LOAD}}}{0.9 E_{unreg}}$$

$$\star I_{DC} = D \times I_L$$

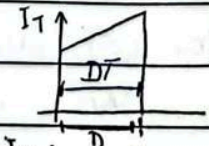


$$\star \Rightarrow I_{DC, \text{LOAD}} = C_{out} \times \frac{\Delta V_{LOAD}}{\Delta t}$$

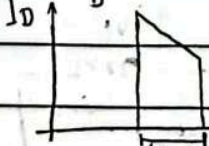
inductor current when transistor is ON.



transistor ON (Transistor current)



(Diode ON) (1-D)T (diode current)



(when transistor - OFF, diode ON)

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$$\left. \begin{aligned} I_{avg, Q} &= D \times I_{DC, I/P} \\ I_{RMS, Q} &= \sqrt{D} \times I_{DC, I/P} \end{aligned} \right\} \text{transistor}$$

$$\begin{aligned} I_{avg, D} &= (1-D) \times I_{DC, I/P} \\ I_{RMS, D} &= \sqrt{(1-D)} \times I_{DC, I/P} \end{aligned}$$

O/p higher than I/p (always)

when transistor is OFF:

$$-E_{unreg} - V_L + V_{LOAD} = 0$$

$$* \quad V_{LOAD} = V_L + E_{unreg}$$

*. keep $\Delta V_{LOAD} < 20 m V_{pp}$ (if ΔV_{pp} is not given)

$$* \quad V_L = \frac{L \Delta i}{(1-D) T} \quad (i) \text{ (discharging)}$$

$$* \quad E_{unreg} = \frac{L \Delta i}{DT} \Rightarrow L = \frac{E_{unreg} \times DT}{\Delta i} \quad (ii)$$

$$\Rightarrow (i) \quad \frac{V_L \times (1-D) T}{L} = \frac{E_{unreg} \times DT}{\Delta i} \Rightarrow (ii) \text{ in } (i)$$

$$V_L = E_{unreg} \times \frac{DT}{\Delta i} \times \frac{\Delta i}{(1-D) T}$$

$$V_L = E_{unreg} \times \frac{D}{(1-D)}$$

$$\begin{aligned} V_{LOAD} &= E_{unreg} + V_L \\ &= E_{unreg} + E_{unreg} \left(\frac{D}{1-D} \right) \end{aligned}$$

$$* \quad V_{LOAD} = E_{unreg} \left(1 + \frac{D}{1-D} \right) = E_{unreg} \left(\frac{1}{1-D} \right) \quad \because D < 1$$

$$\downarrow$$

$$\frac{1-D+D}{1-D}$$

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$$D = \begin{cases} \text{for buck regulator} = V_o/V_{in} \\ \text{for boost regulator} = 1 - \frac{V_{in}}{V_o} \end{cases}$$

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$$C_{OUT} = \frac{I_{DC, LOAD} \times \Delta t}{\Delta V_{LOAD}} = \frac{I_{DC, LOAD} \times DT}{\Delta V_{LOAD}}$$

$$\Delta V_{PP} = \Delta I_{PP, inductor} \times ESR_{C_{OUT}} \quad , \quad C_{OUT} = 1500 \mu F \times \frac{I_{DC, LOAD}}{A}$$

EXAMPLE # 7.6 (Boost regulator $\rightarrow O/P > I/P$)

$$V_{in} = 12 V_{DC} \quad , \quad V_{LOAD} = 22 V_{DC} \quad \& \quad 0.7 A_{DC} = I_{LOAD}$$

$$f = 100 kHz \Rightarrow T = 1/100k = 1 \times 10^{-5}$$

$$R_{ON} = 0.4 \Omega \quad , \quad \text{diode drops} = 0.5 V$$

$$i) P_{LOAD} = P_{in} \times 0.9$$

$$= (12 \times 1.425) \times 0.9$$

$$P_{LOAD} = 15.39 W \approx 15.4 W$$

$$I_{DC, I/P} = \frac{22 \times 0.7}{0.9 \times 12} = 1.425$$

$$ii) I_{DC, I/P} = 1.425 A$$

$$iii) \Delta I_{PP} = 0.35 \quad (25\% \text{ of } I_{DC, I/P})$$

$$\therefore 0.25 \times 1.4 = 0.35$$

$$\Delta t = DT$$

$$L = \frac{E_{unreg} \times \Delta t}{\Delta I_{PP}} = \frac{12 \times (0.45)(1 \times 10^{-5})}{0.35}$$

$$* D = 1 - \frac{E_{unreg}}{V_{LOAD}} = 0.45$$

$$L = 3.08 \times 10^{-4} \times 1.54 \times 10^{-4} = 154 \mu H$$

$$iv) \text{Current rating} = I_{L, P} = \frac{1.4 + 0.35}{2}$$

$$= 1.6 A_P$$

$$v) P_R = D \times I_L^2 \times R_{ON} = R_{ON}$$

$$= 0.45 \times 1.4^2 \times 0.4$$

$$= 353 mW$$

* Buck $\rightarrow I/p > o/p$
 Boost $\rightarrow I/p < o/p$

* for C_{out} , use $20m$ if ΔV_{pp} is not given

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vii) $P_{DIODE} = (1-D) \times I_L \times V_D = (1-0.45) \times 1.4 \times 0.5$
 $= 390 \text{ mW}$

viii) $C_{out} = \frac{0.7 \times 0.45 \times 1 \times 10^{-5}}{20m \rightarrow \Delta V_L \leq 20m}$
 $= 1.575 \times 10^{-9} = 158 \mu F$

16/8 BOOST REGULATOR LM2585 ($o/p > I/p$)

• $V_{DC, LOAD} = 1.23 V_{DC} \left(\frac{1 + R_f}{R_i} \right)$; $1k\Omega \leq R_i \leq 5k\Omega$

• $P_Q = I_{RMS,Q}^2 \times R_{ON,Q}$; $I_{RMS,Q} = \sqrt{D} I_{DC, LOAD}$; $R_{ON,Q} = 0.15\Omega$

• $P_{IC} = \frac{1}{50} \times D \times I_{DC, pp} \times E_{unreg}$

• Total power = $P_Q + P_{IC}$

EXAMPLE # 7.7

$E_{unreg} = 12 V_{DC}$, $V_{LOAD} = V_O = 22 V_{DC}$, $I_{LOAD} = 0.7 A_{DC}$
 $T_A = 80^\circ C$

i) $V_{DC, LOAD} = (1.23 V_{DC}) \left(\frac{1 + R_f}{R_i} \right)$

$\frac{R_f}{R_i} = \frac{22 V_{DC} - 1}{1.23 V_{DC}}$

$\frac{R_f}{R_i} = 16.9$ (i)

Assume, $R_i = 1.3k\Omega$

$\Rightarrow$ (i) $R_f = 16.9 \times 1.3k = 21.9k\Omega \approx 22k\Omega$

EXAMPLE #2 (PRACTICE PROBLEM)

= BOOST REGULATOR.

x $E_{unreg} = 22 \text{ V}_{DC}$, $I_{DC, I/P} = 0.7 \text{ A}_{DC}$

- $V_o = V_{LOAD} = 56 \text{ V}_{DC}$

- $f = 100 \text{ KHz} \Rightarrow T = 1 \times 10^{-5} \text{ s}$

- $R_{ON} = 0.4 \Omega$, $V_D = 0.5 \text{ V}$

i) $P_{LOAD} = P_{in} \times 0.9 = V_{in} \times I_{DC, I/P} \times 0.9$
 $= 22 \times 0.7 \times 0.9$
 $= 13.86 \text{ W}$

ii) $\Delta i_{pp} = 0.25 \times 0.7$
 $= 0.175 \text{ A}$

$$\Delta t = DT$$
$$= 6.07 \mu\text{s}$$

$$\therefore D = 1 - \frac{V_{in}}{V_{out}}$$
$$= 0.607$$

$$L = \frac{V_{in} \times \Delta t}{\Delta i_{pp}} = \frac{22 \times 6.07 \mu\text{s}}{0.175}$$

$$L = 763 \mu\text{H}$$

$$\text{Current ratings} = I_{L, \text{PEAK}}$$

$$= I_{LP} + \frac{\Delta I_{PP}}{2}$$

$$= 0.7 + \frac{0.175}{2} = 0.7875$$

$$I_{DC, LP} = I_L$$

$$P_Q = D \times I_L^2 \times R_{ON} = 0.607 \times (0.7)^2 \times 0.4$$

$$= 119 \text{ mW}$$

$$P_D = (1-D) \times I_L \times V_D$$

$$= (1-0.607) \times (0.7) \times (0.5)$$

$$= 137.5 \text{ mW}$$

$$C_{out} = \frac{I_{DC, LOAD} \times \Delta t}{\Delta V_{LOAD}} \quad (i)$$

$$I_{DC, LOAD} = \frac{P_{LOAD}}{V_{LOAD}} = \frac{13.86}{56} = 0.2475$$

$$i) C_{out} = \frac{0.2475 \times 6.07 \mu}{20 \text{ m}}$$

$$= 75.1 \mu\text{F}$$

Date: _____

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$$P_{im} = (12 \times 1.425) \\ = 17.1 \text{ W}$$

$$I_{DC}, I_{pp} = 1.425 \text{ A}$$

$$P_Q = \frac{(\sqrt{D} I_{DC, I_{pp}})^2 \times R_{ON, Q}}{2} \\ = \frac{(\sqrt{0.45} (1.4))^2 (0.15)}{2} \\ = 0.1323 \text{ W} = 132 \text{ mW}$$

$$P_{IC} = \frac{1}{50} \times 0.45 \times 1.4 \times 12 \\ = 0.1512 = 151 \text{ mW}$$

$$P_T = 132 \text{ m} + 151 \text{ m} \\ = 283 \text{ mW}$$

$$T_J = 80 + (0.283)(65^\circ) \quad \theta_{JA} \\ = 98.4^\circ \text{C} < 110^\circ \text{C} \quad (\text{no heat sink})$$

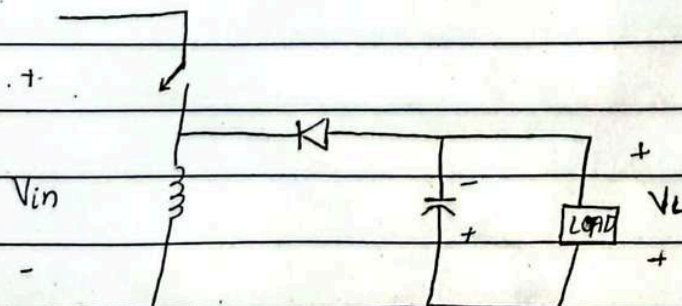
$$V_o = D \times V_{in} \quad \rightarrow \quad V_o = \frac{1}{1-D} \times V_{in}$$

BUCK-BOOST CONVERTER:

$$V_o = \frac{D}{1-D} \times V_{in}$$

* $D > 0.5$ (50%) (Boost converter)

* $D < 0.5$ (Buck converter)



$$V_L = V_{in} D T - V_o (1-D) T$$

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EXAMPLE #4

(PROBLEM #7.3)

(Buck regulator)

$$E_{inreg} = 9V_{DC}, V_{LOAD} = 5V_{DC}, I_{DC, LOAD} = 0.8A,$$

$$f = 150kHz, \Delta V_{PP} = 0.1V$$

$$\Rightarrow T = 6.67\mu s$$

$$D = \frac{5}{9} = 0.56$$

$$\Delta t = D \times T = (0.56)(6.67\mu s) = 3.73\mu s$$

$$\Delta i_{PP} = \cancel{0.4938A} \left(\frac{I_{DC, LOAD}}{V_L} \right) = 0.2$$

$$\rightarrow \text{OR } (0.8)(0.25) = 0.2$$

$$I_{DC, I/P} = \frac{5 \times 0.8}{0.9 \times 9} = 0.4938 = 493.8m$$

$$\cancel{L = 9 \times 3.73\mu s}$$

$$V_L = L \frac{\Delta i}{\Delta t}$$

$$V_L = 9 - 5 = 4$$

$$L = \frac{V_L \times \Delta t}{\Delta i_{PP}} = \frac{4(3.73\mu s)}{(0.8/4)} = 74.6\mu H$$

$$C_{out} = 500\mu \times 0.8 = 400\mu F$$

$$ESR_{out} = \frac{\Delta V_{PP}}{\Delta i_{PP}} = \frac{0.1}{0.2} = 0.5$$

$$P_{inreg} = P_{in} = 9 \times 0.49 = 4.41W$$

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EXAMPLE #5 (PROBLEM # 7.11) (Boost regulator)

$$f = 100 \text{ kHz}, E_{\text{unreg}} = 5 \text{ V}_{\text{DC}}, V_{\text{LOAD}} = 9 \text{ V}_{\text{DC}}, I_{\text{DC, LOAD}} = 0.8 \text{ A}_{\text{DC}},$$

$$T = 1 \times 10^{-5} \text{ s}$$

$$\Delta V_{\text{PP}} = 0.1 \text{ V}$$

$$\Delta t = 0.44 \times 1 \times 10^{-5}$$

$$= 4.4 \mu\text{s}$$

$$D = \frac{1-5}{9} = 0.44$$

$$\Delta i_{\text{PP}} = 0.25 \times I_{\text{DC, I/P}}$$

$$= 1.6 \times 0.25$$

$$= 0.4$$

$$I_{\text{DC, I/P}} = \frac{9 \times 0.8}{0.9 \times 5} = 1.6$$

$$L = \frac{E_{\text{unreg}} \times \Delta t}{\Delta i_{\text{PP}}} = \frac{5 \times 4.4 \mu}{0.4}$$

$$L = 55 \mu\text{H}$$

$$C_{\text{out}} = \frac{0.8 \times 4.4 \mu}{\Delta V_{\text{PP}}} = 36 \mu\text{F}$$

$$P_{\text{unreg}} = P_{\text{in}} = 5 \times 1.6$$

$$= 8 \text{ W}$$

* Diode do not conduct in +ve

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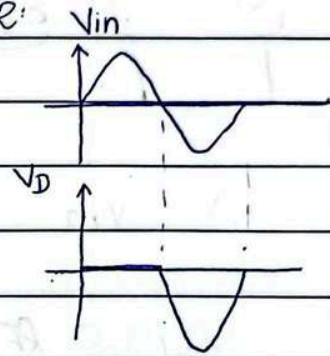
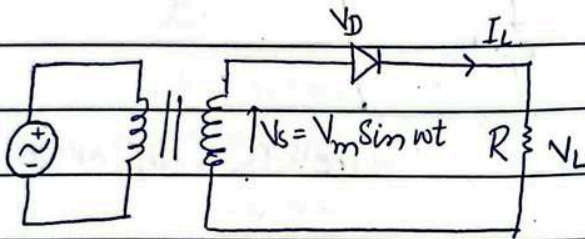
CHAPTER #10

(By ^{from} power electronic converters by Hamoon Rashid)

UNCONTROLLED RECTIFIERS:

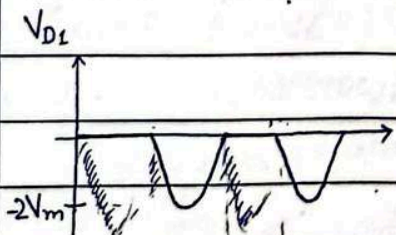
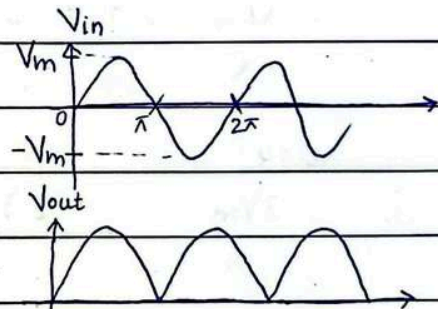
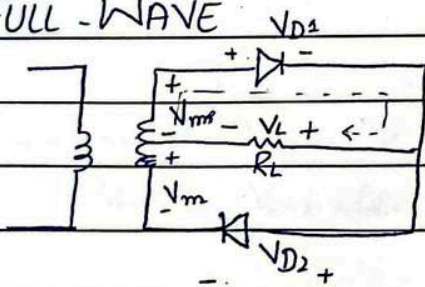
Anode more +ve - diode ON.

SINGLE-PHASE HALF WAVE RECTIFIER:



Peak inverse Voltage = PIV = V_m

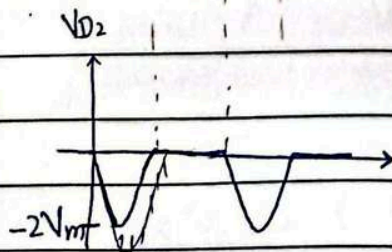
FULL-WAVE



$$V_m + V_m = 2V_m$$

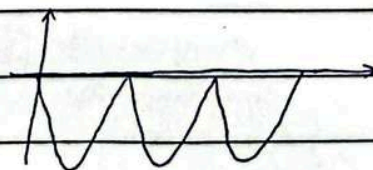
$$V_{D1} + V_L$$

* PIV = $2V_m$ ($V_m + V_m$)



$$V_{D1} + V_L + V_m = 0$$

$$V_D = V_{D1} + V_{D2}$$

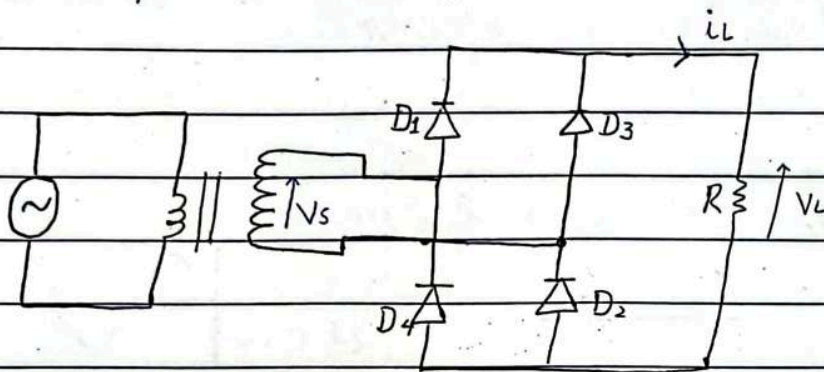


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SINGLE-PHASE BRIDGE RECTIFIER



* $PIV = V_m$ (Same as half wave)

VOLTAGE RELATIONSHIPS

$$* V_{DC} = \frac{1}{T} \int_0^T v(t) dt$$

→ AVERAGE VALUE:

• Half wave:

$$V_{DC} = V_{avg} = \frac{V_m}{\pi} = 0.318 V_m$$

• full wave:

$$V_{DC} = V_{avg} = \frac{2V_m}{\pi} = 0.636 V_m$$

→ RMS VALUE:

• ~~Half~~ Half wave:

$$V_{rms} = \frac{V_m}{2} = 0.5 V_m$$

• Full wave:

$$V_{rms} = \frac{V_m}{\sqrt{2}} = 0.707 V_m$$

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CURRENT RELATIONSHIPS

- $I_{DC} = \frac{V_{DC}}{R}$ (Half wave = Full wave)
- $I_{RMS} = \frac{V_{RMS}}{R}$ (Half wave = Full wave)

OUTPUT DC POWER:

- $P_{DC} = V_{DC} \times I_{DC}$

OUTPUT AC POWER

- $P_{AC} = V_{RMS} \times I_{RMS}$

RECTIFICATION RATIO (CONVERSION EFFICIENCY)

* • $\eta = \frac{P_{DC}}{P_{AC}} = \frac{V_{DC} \times I_{DC}}{V_{RMS} \times I_{RMS}}$

HALF WAVE:

$$\begin{aligned} \eta &= \frac{V_{DC} \times (V_{DC}/R)}{V_{RMS} \times (V_{RMS}/R)} = \frac{V_{DC}^2}{V_{RMS}^2} = \frac{(V_m/\pi)^2}{(2V_m/\pi)^2} \\ &= \frac{V_m^2}{\pi^2} \times \frac{4}{V_m^2} = \frac{4}{\pi^2} = 0.405 = 40.5\% \end{aligned}$$

FULL WAVE:

$$\begin{aligned} \eta &= \frac{V_{DC}^2}{V_{RMS}^2} = \frac{(2V_m/\pi)^2}{(V_m/\sqrt{2})^2} = \frac{4V_m^2}{\pi^2} \times \frac{2}{V_m^2} \\ &= \frac{8}{\pi^2} = 0.81 = 81\% \end{aligned}$$

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FORM FACTOR (FF)

$$FF = \frac{V_{RMS}}{V_{DC}}$$

HALF WAVE:

$$FF = \frac{V_m/2}{V_m/\pi} = \frac{\pi}{2} = 1.57$$

FULL WAVE:

$$FF = \frac{V_m/\sqrt{2}}{2V_m/\pi} = 1.11$$

RIPPLE FACTOR:

$$RF = \frac{V_{AC}}{V_{DC}} \quad (i)$$

$$\therefore V_{AC} = \sqrt{V_{RMS}^2 - V_{DC}^2}$$

$$\Rightarrow (i) \quad RF = \frac{\sqrt{V_{RMS}^2 - V_{DC}^2}}{\sqrt{V_{DC}^2}} = \sqrt{\frac{V_{RMS}^2}{V_{DC}^2} - 1}$$

$$= \sqrt{FF^2 - 1}$$

TRANSFORMER UTILIZATION FACTOR:

$$TUF = \frac{P_{DC}}{V_s I_s} = \frac{V_{DC} \times I_{DC}}{V_s \times I_s}$$

$$\therefore V_s = \frac{V_m}{\sqrt{2}} = 0.707 V_m$$

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$$* I_s = I_L$$

$$\Rightarrow I_s = \frac{0.5 V_m}{R} \quad , \quad I_s = \frac{0.707 V_m}{R}$$

• HALF-WAVE:

$$TUF = \frac{(V_m/\pi)(V_{DC}/R)}{(0.707 V_m)(0.5 V_m/R)} \quad V_{DC} = \frac{V_m}{\pi}$$

$$= \frac{V_m^2/\pi^2}{0.3535 V_m^2} = \frac{1}{0.3535 \pi^2}$$

$$TUF = 0.286$$

• FULL-WAVE:

for center tapped;
full wave $V_s I_s$ is $\leftarrow 2 \times V_s \times I_s$
double of half wave-
rectifiers or 2 half
wave rectifiers are
operating together.

$$TUF = P_{DC} = \frac{(2V_m/\pi)(2V_m/\pi R)}{2 \times (V_m/\sqrt{2})(0.5 V_m/R)}$$

$$= \frac{4V_m^2/\pi^2 R}{2 \times (0.5 V_m^2)/\sqrt{2} R}$$

$$= \frac{4}{\pi^2} \times \frac{\sqrt{2}}{0.5} \times \frac{1}{2} = 0.572$$

• BRIDGE RECTIFIER:

$$TUF = \frac{P_{DC}}{V_s I_s} = \frac{V_{DC} \times I_{DC}}{V_s \times I_s}$$

$$= \frac{(2V_m/\pi)(2V_m/\pi R)}{(0.707 V_m)(0.707 V_m/R)} = \frac{4/\pi^2}{(0.707)^2}$$

$$TUF = 0.81$$

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POWER FACTOR:

$$PF = \frac{P_{AC}}{V_S \times I_S}$$

CREST FACTOR:

$$CF = \frac{I_S (\text{PEAK})}{I_S}$$

(measure of peak I/P current compared to RMS of I_L).

EXAMPLE # 1

Full wave transformer - center tapped:

$$\begin{aligned} \text{i) Efficiency} = \sigma &= \frac{P_{DC}}{P_{AC}} = \frac{V_{DC} \times I_{DC}}{V_{RMS} \times I_{RMS}} \\ &= \frac{(2V_m/\pi)(2V_m/\pi R)}{(0.707 V_m)(0.707 V_m/R)} \\ &= \frac{4/\pi^2}{(0.707)^2} \end{aligned}$$

$$\sigma = 0.81$$

$$\begin{aligned} \text{ii) Form Factor} = FF &= \frac{V_{RMS}}{V_{DC}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} \\ &= \frac{V_m}{\sqrt{2}} \times \frac{\pi}{2V_m} \\ FF &= 1.11 \end{aligned}$$

$$\begin{aligned} \text{iii) Ripple factor} = RF &= \sqrt{FF^2 - 1} \\ &= \sqrt{(1.11)^2 - 1} \\ &= 0.48 \end{aligned}$$

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iv) $TUF = 0.572$

v) $PIV_{D1} = 2V_m$

(PIV is same on both diodes $\Rightarrow 2V_m$)

$PIV_{FW} = 2V_m$

vi) $CF = \frac{I_{S(Peak)}}{I_S}$

$RMS = \frac{PEAK}{\sqrt{2}}$

$I_{S(Peak)} = \frac{\sqrt{2} \times 0.707 V_m}{R}$

$I_{S(FW)} = \frac{0.707 V_m}{R}$

$I_{S(Peak)} = \frac{0.999 V_m}{R}$

$\Rightarrow CF = \frac{0.999 V_m / R}{0.707 V_m / R} = 1.414$

vii) $PF = \frac{P_{AC}}{V_S \times I_S} = \frac{V_{RMS} \times I_{RMS}}{V_S \times I_S}$

$= \frac{(V_m / \sqrt{2}) (V_m / \sqrt{2} R)}{(0.707 V_m) (0.707 V_m / R)}$

$= \frac{1/(\sqrt{2})^2}{(0.707)^2}$

$= \frac{1/(\sqrt{2})^2}{(0.707)^2}$

$PF = 1$

23/4 EXAMPLE # 2

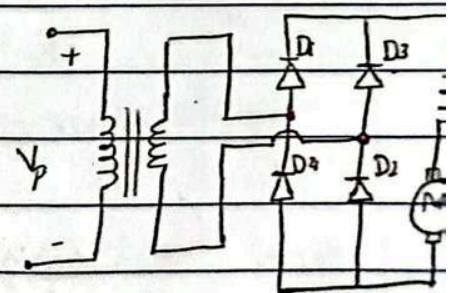
Single phase bridge rectifier

$TUF = 1$, $I_{S1} = 0.9 I_a$

$I_S = I_a$, $\phi = 0^\circ \Rightarrow \cos 0^\circ = 1$

$HF = ?$, $PF = ?$

$HF = T^* HD$



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$$* \quad HIF = \left(\frac{I_s^2 - I_{s1}^2}{I_{s1}^2} \right)^{1/2} = \left[\left(\frac{I_s}{I_{s1}} \right)^2 - 1 \right]^{1/2}$$

$$= \sqrt{\left(\frac{I_a}{0.9 I_a} \right)^2 - 1} = 0.484$$

$$* \quad PF = \frac{I_{s1} \cos \phi}{I_s} = \frac{0.9 I_a (\cos 0^\circ)}{I_a}$$

$$= 0.9 \quad (\text{lagging})$$

* $\left[\begin{array}{l} \rightarrow \text{inductive load} - \text{lagging} \\ \rightarrow \text{capacitive load} - \text{leading} \end{array} \right.$

DESIGN CONSIDERATIONS:

If, $V_{DC} = 10V$

For Half wave:

$$i) \quad V_s: \quad \left. \begin{array}{l} V_{DC} = 0.318 V_m \\ V_{RMS} = 0.5 \cancel{10} V_m \end{array} \right\} \text{O/P} \quad \Rightarrow V_m = V_{DC} / 0.318$$

$$* \quad V_{s, \text{I/p secondary voltage}} = V_s = 0.707 V_m$$

$$V_{s, \text{H.W}} = 0.707 \frac{V_{DC}}{0.318}$$

$$= 2.22 V_{DC}$$

$$V_{s, F.W} = 0.707 \frac{V_{DC}}{0.636}$$

$$V_{DC, F.W} = 0.636 V_m$$

$$= 1.11$$

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ii) PIV: $\nearrow$ Peak repetitive reverse voltage

→ Half wave: $V_{RRM} = V_m = \frac{V_{DC}}{0.318} = 3.14 V_{DC}$

$$\therefore PIV_{H.W} = V_m$$

→ Full wave

$$\bullet \text{ Center tapped } = V_{RRM} = \frac{2V_m}{0.636} \times 2 = \frac{2 \times 1.57}{1.57} V_{DC} = 3.14 V_{DC}$$

$$PIV_{F.W} = 2V_m$$

(center tapped)

$$\bullet \text{ Bridge rectifier } = V_{RRM} = V_m = \frac{V_{DC}}{0.636} = 1.57 V_{DC}$$

$$PIV(\text{bridge rectifier}) = V_m$$

iii) Peak repetitive forward current = $I_{FRM} = V_m/R$

→ Half wave: $I_{DC} = V_{DC}/R \Rightarrow I_{DC} \times R$

$$I_{FRM} = \frac{V_{DC}}{0.318} = \frac{I_{DC} \times R}{0.318 R}$$

$$= 3.14 I_{DC}$$

→ Full wave:

$$I_{FRM} = \frac{V_{DC}}{0.636 R} = 1.57 I_{DC}$$

THREE-PHASE UNCONTROLLED RECTIFIER:

- Single phase — used for low to medium power application
- For output higher than 15kW, three phase rectifiers are used.

TYPES:

- Star rectifier
- Bridge rectifier

* Full wave - conduction angle = 360

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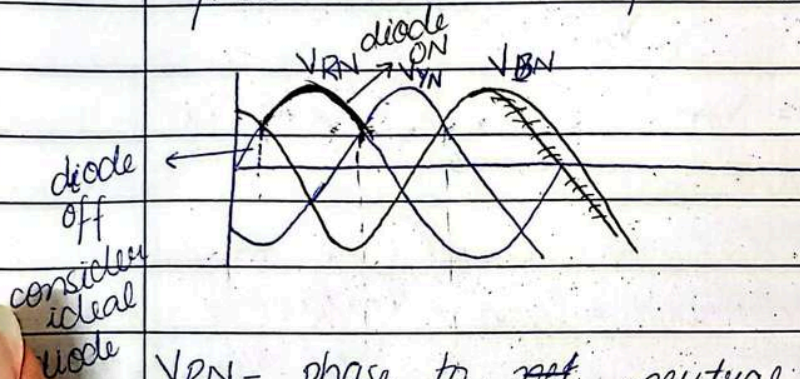
→ STAR RECTIFIERS (Half wave)

• Diode - ON: when voltage on that phase is greater than the other two.

• Diode conduction angle: $2\pi/3$ instead of π .

→ 1 phase → 1 diode ⇒ half wave

→ 2 phase → 2 diode ⇒ full wave.



$$V_m \sin \omega t \text{ dwt}$$

$$= V_m \sin \theta \text{ d}\theta$$

$$T_{ON} = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3} \text{ rad}$$

$$\text{Conduction angle} = 120^\circ \text{ OR } \frac{2\pi}{3} \text{ rad}$$

$$V_{DC} = \frac{1}{T_{ON}} \int_{\pi/6}^{\frac{5\pi}{6}} V_m \sin \theta \text{ d}\theta$$

$$= \frac{3}{2\pi} \int_{\pi/6}^{\frac{5\pi}{6}} V_m \sin \theta \text{ d}\theta$$

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$$V_{DC} = 0.8269 V_m$$

$$\approx 0.827 V_m$$

$$V_{RMS} = \sqrt{\frac{3}{2\pi} \int_{\pi/6}^{5\pi/6} V_m^2 \sin^2 \theta d\theta}$$

$$RMS = \frac{PEAK}{\sqrt{2}}$$

$$V_{RMS} = 0.84 V_m$$

$$I_s = \sqrt{\frac{1}{2\pi} \int_{\pi/6}^{5\pi/6} I_m^2 \sin^2 \theta d\theta}$$

Is works one time for each winding

$$I_s = 0.485 I_m$$

$$I_s = \frac{0.485 V_m}{R}$$

EXAMPLE #3

A three ^{phase} star rectifier:

$$i) \text{ Efficiency} = \eta = \frac{V_{DC} \times I_{DC}}{V_{RMS} \times I_{RMS}}$$

$$= \frac{0.827 V_m \times 0.827 V_m / R}{0.84 V_m \times 0.84 / R}$$

$$I_{RMS} = V_{RMS} / R$$

$$= \frac{0.827 \times 0.827}{0.84 \times 0.84} = 96.9\%$$

$$ii) FF = \frac{V_{RMS}}{V_{DC}} = 1.015$$

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$$\text{iii)} \quad RF = \sqrt{FF^2 - 1} = \sqrt{(1.01)^2 - 1} \\ = 0.141$$

$$\text{iv)} \quad TUF = \frac{P_{DC}}{3 \times V_s \times I_s} = \frac{V_{DC} \times I_{DC}}{3 \times V_s \times I_s}$$

$$V_s = 0.707 V_m$$

$$I_s = 0.485 I_m$$

$$TUF = (0.827)^2 / (3 \times 0.707 \times 0.485)$$

$$TUF = 0.66 < 1 \quad \text{(always less than 1)}$$

$$V_{ab} = V_a - V_b$$

Phase to phase voltage
OR
line-to-line voltage
(line voltage) V_L

Phase-to-neutral voltage (Phase voltage) V_P
OR
line-to-neutral voltage

$$* \quad V_L = \sqrt{3} V_P \quad (140 - 14120 = \sqrt{3})$$

$$\Rightarrow^+ \quad PIV = \sqrt{3} V_m$$

line rating

$$\rightarrow v) \quad PF = \frac{P_{AC}}{3 \times V_s \times I_s} = \frac{0.842}{3 \times 0.707 \times 0.485} \\ = 0.685$$

$$vi) \quad PIV = V_m$$

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vii) $I_{DC} = 30 A$ (given) , $V_{DC} = 140 V$ (given)

$$I_{Diode (avg)} = \frac{30}{3} = 10 \quad (i)$$

$$I_{DC} = 0.827 I_m$$

$$I_{Diode (avg)} = 0.275 I_m \quad (ii) \therefore 0.827/3$$

Compare eq. (i) & (ii)

$$10 = 0.275 I_m$$

$$I_m = 36.27 A \quad (Peak)$$

30/4 THREE PHASE BRIDGE RECTIFIERS:

- Conduction angle = $\frac{2\pi}{3}$

- highest possible TUF

→ No. of diode shows how it will ON.

→ At a time, 2 diodes will be ON.

- Line voltage = $\sqrt{3} V_m$

$$V_p = \sqrt{3} V_m = 1.73 V_m$$

- * $V_{DC} = V_{AVG} = \frac{6}{2\pi} \int_{\pi/3}^{2\pi/3} \sqrt{3} V_m \sin \theta d\theta \quad (6 \text{ waves})$

$$= 1.6539$$

$$= 1.654 V_m$$

- * $V_{RMS} = \sqrt{\frac{6}{2\pi} \int_{\pi/3}^{2\pi/3} (\sqrt{3} V_m \sin \theta)^2 d\theta}$

$$= 1.655 V_m$$

$$I_{s(AVG)} = \frac{4}{2\pi} \int_{\pi/3}^{2\pi/3} I_m \sin \theta d\theta$$

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$$I_s = \frac{4}{2\pi} \int_{\pi/3}^{2\pi/3} I_m^2 \sin^2 \theta d\theta$$

(in transformer:
current flows only if
related diodes are ON)

$$I_s = 0.78 I_m \quad (\text{each winding - transformer secondary current})$$

$$I_m = \frac{\sqrt{3} V_m}{R}$$

$$\text{Diode current} = \frac{\sqrt{3} V_m}{R} I_{D_{RMS}} = \frac{1}{\sqrt{2}} \int_{\pi/3}^{2\pi/3} I_m^2 \sin^2 \theta d\theta$$

$$= 0.552 I_m$$

$$TUF = \frac{(1.654)^2}{3 I_s V_s}$$

$$= \frac{(1.654)^2 (1.654 V_m / R)}{3 \times 0.707 V_m \times 0.78 \times \sqrt{3} V_m / R}$$

$$= 0.954$$

EXAMPLE #4

(from Power electronics 4th Edition
by Hameed Rashid)

$$V_{DC} = 280.7 \text{ V}$$

$$(i) \& \quad V_{DC} = 1.654 V_m \quad (iii)$$

from
(i) & (iii)

$$\Rightarrow V_m = 169.71 \text{ V}$$

$$V_m = \frac{280.7}{1.654}$$

$$PIV = \sqrt{3} V_m$$

$$= 293.94$$

* if not given, consider it as RMS value.

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EXAMPLE #5 (Line to neutral voltage)

$$I_{DC} = 60A$$

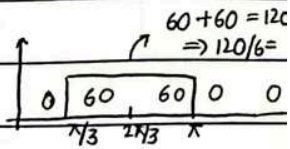
$$V_{supply} = 120$$

$$\rightarrow I_D (avg) = \frac{1}{2\pi} \int_{\pi/3}^{\pi} I_{DC} d\theta = 20A$$

$$\rightarrow I_D (RMS) = \sqrt{\frac{1}{2\pi} \int_{\pi/3}^{\pi} 60^2 d\theta}$$

$$= 34.64$$

$$OR \frac{I_{DC}}{\sqrt{3}} = \frac{60}{\sqrt{3}} = 34.64$$



$$\rightarrow PIV = \sqrt{3} V_m$$

$$= \sqrt{3} \times 120_{RMS}$$

$$= \sqrt{3} \times \sqrt{2} \times 120$$

$$= 293.93$$

$$V_{RMS} = \frac{V_{peak}}{\sqrt{2}}$$

EXAMPLE #6

$$a) I_{RMS} = ?$$

$$t_1 = 100\mu s, t_2 = 350\mu s, t_3 = 500\mu s,$$

$$f = 250Hz, f_s = 5kHz, I_m = 450A, I_a = 150A$$

$$I_{(AVG)} = \frac{1}{T} \int_0^{t_1} I_m \sin \omega_0 t dt + \frac{1}{T} \int_{t_2}^{t_3} i_2 dt$$

$\swarrow \frac{1}{T} = 250$
 $\swarrow \frac{250}{\pi} = 5000$
 $\swarrow i_2 = I_a$

$$\therefore \omega_0 = 2\pi f_s = (2\pi \times 5000)$$

$$= \frac{1}{1000} 250 \left[\int_0^{100\mu} 450 \sin(2\pi \times 5000 t) dt + \int_{350\mu}^{500\mu} 150 dt \right]$$

$$= 12.79$$

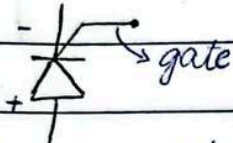
$$I_{RMS} = 58.09$$

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THYRISTORS

(~~un~~controlled rectifiers)



(it will not work until trigger is

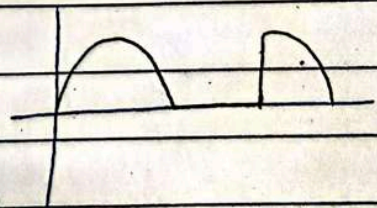
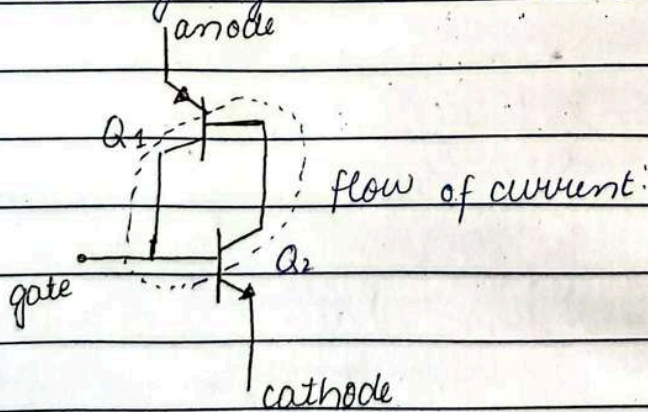
(will also work if applied at gate)

voltage is increased - No gate pulse applied)

- Delivers 1000's of W to the load of 98% efficiency.

→ SCR (Silicon controlled rectifiers)

- Circuits that protect thyristors and line are called snubbers.
- Q_2 keeps Q_1 ON, Q_1 keeps Q_2 ON.
- Thyristors - 4 layers $\Rightarrow$ PNP
- Inherently they are always OFF.
- To stop working thyristors - reduce its current.



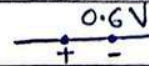
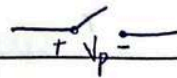
5/5 SILICON CONTROLLED RECTIFIER

- Current above holding current $\Rightarrow$ SCR - ON
 I_H

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- $V_{TM(min)} = 0.6V$
- I_{SM} : Surge current
Thousands of Watts delivers to load.



- $I_{RMS} = I_P/2$
- V_{gt} :
0.7V to forward-bias the gate collector junction.

SNUBBING

- Thyristors turn on in $10\mu s$ to $100\mu s$.
- Voltage across them $\downarrow$ (line peak of 100's to 2V)
- Current increases from 0 to tens to hundreds Amperes
- to control di/dt place an inductor in series with the thyristor.
- in snubbers: (CRITICAL RATE OF RISE ^{OF} ~~I~~ CURRENT)

$$* \quad L > \frac{V_{P, LINE (peak)}}{(di/dt)_{(peak) spec.}}$$

EXAMPLE #1

$$V_{LINE} = 220V_{RMS}, \quad f_{LINE} = 50Hz$$

$$R_L = 80\Omega, \quad di/dt = 50A/\mu s.$$

$$a) \quad L > \frac{220\sqrt{2}}{50/\mu s} = 4.4\mu H$$

$$L > 6.22\mu H$$



Date: _____

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b) $Z_L = j\omega L$ (impedance of inductor)
 $= (2\pi (50)) (6.2 \mu) \angle 90^\circ$
 $\rightarrow +j$
 $= 1.95_m \angle 90^\circ$
 $= 0.00195 \angle 90^\circ$

CRITICAL RATE OF RISE OF VOLTAGE:

- Thyristor may turn on due to noise (w/o gate pulse) because of parasitic capacitance b/w main terminal & gate.

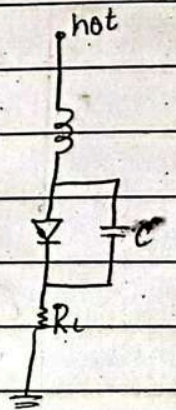
$\rightarrow V_c = V_{p, noise} (1 - e^{-t/(R_L \times C_{snub})})$

$\rightarrow \tau = 1/RC \Rightarrow \text{natural response} = V_0 e^{-t/\tau} = V_0 e^{-t/RC}$

$\rightarrow \frac{dV_c(snub)}{dt} = \frac{V_{p, noise}}{R_L \times C_{snub}} e^{-t/(R_L \times C_{snub})}$

$\rightarrow$ Worst case, $t=0$ capacitor charges quickly.

$\frac{dV_c(snub)}{dt} \bigg|_{\text{worst case, } t=0} = \frac{V_{p, noise}}{R_L \times C_{snub}}$



$\rightarrow i = C \frac{dV_c}{dt}$

* Use V_{RMS} as $V_{p, noise}$

$C_{snub} > \frac{V_{DRM}}{R_{LOAD} \times (dV/dt)_{spec}}$

$R_{LOAD} \times (dV/dt)_{spec}$

data by manufacturer

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EXAMPLE #2

$$V_{LINE} = 220 V_{RMS}, \quad f_{LINE} = 50 Hz, \quad V_{DRM} = 400 V_{(PEP)}$$

$$R_L = 80 \Omega, \quad dv/dt = 75 / \mu V$$

$$a) \quad C > \frac{V_{RM}}{R_L \times (dv/dt)} = \frac{400}{80(75/\mu)}$$

$$C > 66.7 nF = 0.067 \mu F$$

$$C \approx 0.1 \mu F$$

$$b) \quad i = \frac{C dv}{dt} = (0.067 \mu)(75/\mu)$$

$$Z_{Xc} = \frac{1}{j\omega C}$$

$$*X_c = 1/\omega C \text{ (w/o } j)$$

$$= \frac{1}{\angle -90^\circ}$$

$$(2\pi \times 50)(0.1 \mu F)$$

$$= 31830.9 \angle -90^\circ$$

$$= 32k \angle -90^\circ$$

$$* I_{Leakage} = I_{LOAD} = \frac{V_{RMS}}{R_{LOAD} + Z_c} = \frac{220}{(80) + (32k \angle -90^\circ)}$$

$$= 6.87_m \angle 89.85^\circ$$

$$= 6.9_m \angle 90^\circ$$

$$* I_L = I_{RMS}$$

$$V_{LOAD} = (6.9_m \angle 90^\circ)(80)$$

$$= 0.5 V_{RMS} \angle 90^\circ$$

$$P_{LOAD} = I_{LOAD} \times V_{LOAD}$$

$$= 3m W$$

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LEAKAGE CURRENT:

- Current through capacitor is leakage current.

- $$I_{\text{discharge}} = \frac{V_{P, \text{LINE}}}{R_{\text{snubbing}}}$$

- Total current at anode becomes.

$$I_{\text{SCR}} = I_{P, \text{LOAD}} + I_{\text{discharge}} < I_{\text{TSM}}$$

$$I_{\text{TSM}} > I_{P, \text{LOAD}} + I_{\text{discharge}}$$

- $$\Rightarrow R_{\text{snubbing}} > \frac{V_{P, \text{LINE}}}{I_{\text{TSM}} - I_{P, \text{LOAD}}}$$

EXAMPLE #3

$$V_{\text{line}} = 220 \text{ VRMS}$$

$$f_{\text{LINE}} = 50 \text{ Hz}$$

$$R_{\text{LOAD}} = 80 \Omega$$

$$I_{\text{TSM}} = 46 \text{ A}$$

$$R_{\text{snubbing}} > \frac{220 \sqrt{2}}{46 - \left(\frac{220 \sqrt{2}}{80} \right)}$$

$\nearrow V_{\text{LINE}}$
 $\searrow R_{\text{LOAD}}$

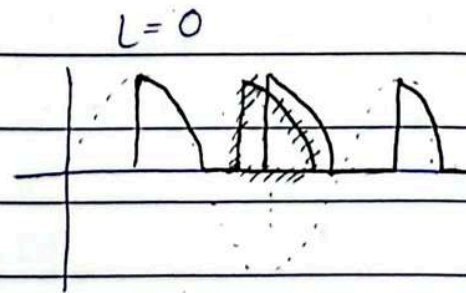
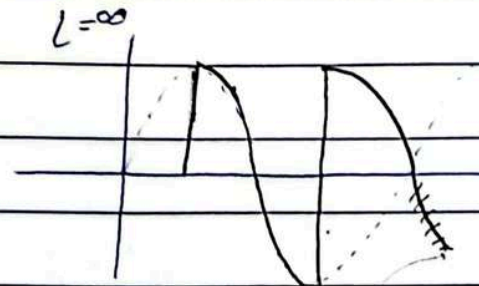
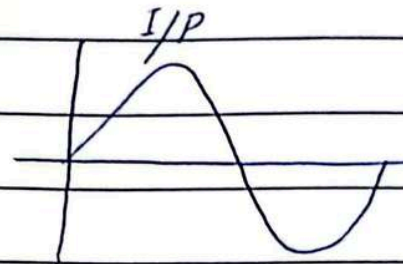
$$= 7.38 \Omega$$

Date: 14/5

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EXAMPLE #

(when load $\rightarrow R \ \& \ L$)



- Three phase half wave rectifiers
→ 3 SCR's

Three phase bridge rectifiers

- 6 SCR's
→ AVERAGE & RMS VALUES

$$V_{an} = V_m \angle 0^\circ = V_m \sin \omega t$$

$$V_{bn} = V_m \angle -120^\circ = V_m \sin \left(\omega t - \frac{2\pi}{3} \right)$$

~~V_{cn}~~

$$V_{cn} = V_m \angle 120^\circ = V_m \sin \left(\omega t + \frac{2\pi}{3} \right)$$

Line-to-neutral
voltages

$$V_{ab} = V_{an} - V_{bn} = \sqrt{3} V_m \angle 30^\circ$$

$$V_{bc} = V_{bn} - V_{cn} = \sqrt{3} V_m \angle -90^\circ$$

$$V_{ca} = V_{cn} - V_{an} = \sqrt{3} V_m \angle 150^\circ$$

0 + 30°
-120 + 30°
120 + 30°
(line-to-line
voltages)

* line to line voltages → lead by $\angle 30^\circ$

- Slide #1 (wave → 3)

$$V_{dc} = \frac{6}{2\pi} \int_{\frac{\pi}{6} + \alpha}^{(\frac{\pi}{2}) + \alpha} \sqrt{3} V_m \sin \omega t \, d\omega t$$

$$= \frac{3}{\pi} \int_{\frac{\pi}{6} + \alpha}^{\frac{\pi}{2} + \alpha} \sqrt{3} V_m \sin \left(\omega t + \frac{\pi}{6} \right) d\omega t$$

Date: _____

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$$= \frac{3\sqrt{3} V_m}{\pi} \left[-\cos \left(\omega t + \frac{\pi}{6} \right) \right]_{\frac{\pi}{6} + \alpha}^{\frac{\pi}{2} + \alpha}$$

$$= \frac{3\sqrt{3} V_m}{\pi} \left[-\cos \left(\frac{\pi}{2} + \alpha + \frac{\pi}{6} \right) + \cos \left(\frac{\pi}{6} + \alpha + \frac{\pi}{6} \right) \right]$$

$$= \frac{3\sqrt{3} V_m}{\pi} \left[\cos \left(\frac{\pi}{3} + \alpha \right) - \cos \left(\frac{2\pi}{3} + \alpha \right) \right]$$

$$= \frac{3\sqrt{3} V_m}{\pi} \left[\cos \frac{\pi}{3} \cos \alpha - \sin \frac{\pi}{3} \sin \alpha - \cos \frac{2\pi}{3} \cos \alpha + \sin \frac{2\pi}{3} \sin \alpha \right]$$

$$= \frac{3\sqrt{3} V_m}{\pi} \left[\frac{1}{2} \cos \alpha - \frac{\sqrt{3}}{2} \sin \alpha + \frac{1}{2} \cos \alpha + \frac{\sqrt{3}}{2} \sin \alpha \right]$$

$$= \frac{3\sqrt{3} V_m}{\pi} \cos \alpha$$

EXAMPLE #

Three-phase Y-connected 208V, 60Hz

$R = 10 \Omega$, $V_o(\text{avg}) = 50\%$

$$V_s = \frac{208}{\sqrt{3}} = 120.1 \text{ V}$$

$$V_m = 120.1 \sqrt{2} = 169.83 \text{ V}$$

$$\rightarrow \text{avg} = \frac{3}{\sqrt{3}}$$

$$\rightarrow \text{RMS} = \frac{3}{\sqrt{3}}$$

Date: _____

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$$V_{dm} = \frac{3\sqrt{3} V_m}{\pi}, \quad V_{dc} = 0.5 V_{dm}$$

$$= 280.9 \text{ V}, \quad V_{dc} = 140.45 \text{ V}$$

$$\frac{V_m}{V_{dm}} = \frac{V_{dc}}{V_{dm}} = 0.5 = \cos \alpha$$

a) $\Rightarrow \alpha = \cos^{-1}(0.5) \Rightarrow \alpha = 60^\circ$

b) $V_{dc} = 140.45 \text{ V}$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{140.45}{10}$$

$$I_{dc} = 14.05 \text{ A} \quad \because \alpha = 60^\circ$$

$$V_{RMS} = \sqrt{\frac{6}{2\pi} \int_{\frac{\pi}{6}}^{(\frac{\pi}{2})+\alpha} (\sqrt{3} V_m)^2 \sin^2 \left(\omega t + \frac{\pi}{6} \right) d\omega t}$$

$$= 159.29 \text{ V}$$

$$I_{RMS} = \frac{V_{RMS}}{R} = 15.93 \text{ A}$$

$$I_{T, \text{avg}} = \frac{V_{RMS}}{3} I_{dc} = \frac{15.93}{3} \times 14.05 = 4.68 \text{ A} \quad \left(\begin{array}{l} \text{Slide \# 11} \\ \text{Wave \# 4} \end{array} \right)$$

$$I_{T, RMS} = \sqrt{\frac{2}{6}} I_{RMS} = \sqrt{\frac{2}{6}} \times 15.93$$

$$I_{T, RMS} = 9.197 \text{ A}$$

Date: _____

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c) $\eta =$

d) $TUF = \frac{V_{DC} \times I_{DC}}{3 \times V_s \times I_s} \quad (i)$

$I_s = \sqrt{\frac{4}{6}} \times I_{RMS} = 13A$

$\Rightarrow (i) TUF = \frac{140.45 \times 14.05}{3 \times 120.1 \times 13} = 0.42 \quad 21$

e) $P_{out} = I_{RMS}^2 \times R = 15.93^2 \times 10 = 2537.6 W$

f) $PF = \frac{\text{Active power}}{\text{Apparent power}} = \frac{P_{out}}{3 \times V_s \times I_s} = \frac{2537.6}{3 \times 120.1 \times 13}$

$PF = 0.541$